

Experiment No. 21 (Mixture No. 2)

Date: / /

Aim : Analyse two acidic (anion) radicals qualitatively from given inorganic mixture.

Apparatus : Test tubes, test tube holder, test tube stand, filter paper etc.

A. Preliminary tests:

Test	Observation	Inference
Color	white/colourless	chloride, bromide, nitrate, sulphates, carbonate of Ca^{2+} , Ba^{2+} , Sr^{2+} or Mg^{2+} may be present
Nature	crystalline (soluble in water)	Cl^- , Br^- , I^- , NO_2^- , SO_4^{2-} , CO_3^{2-} , CH_3^- , COO^- or compounds of NH_3 may be present.

B. Dry Tests for Acidic radicals (Test tube must be dry for following tests)

Test	Observations	Inference
1. Heating in a dry test tube:- Take a small quantity of the mixture in a clean and dry test tube and heat it strongly	Roddish brown gas which turns moist starch paper orange yellow	Br^- may be Present
2. Action of dil. H_2SO_4 : Take a small quantity of mixture + dilute H_2SO_4	No reaction	CO_3^{2-} , S^{2-} , NO_2^- , SO_3^{2-} , CH_3COO^- are absent.
3. Action of conc. H_2SO_4 : Take a small quantity of mixture + conc. H_2SO_4	Yellowish brown gas having pungent smell, turning starch paper yellow	Br^- may be Present
4. Action of Cu foils and conc. H_2SO_4 Mixture + Cu filings and conc. H_2SO_4 , heat it strongly	—	—
5. Action of MnO_2 and conc. H_2SO_4 Mixture + MnO_2 powder + conc. H_2SO_4	Roddish brown gas turning starch paper yellowish brown	Br^- may be Present

Individual dry tests for SO_4^{2-} & PO_4^{3-} (If required)

1. Test for sulphate Mixture + dil. HCl , boil the solution + BaCl_2 solution	white ppt insoluble in conc. HCl	SO_4^{2-} may be present
2. Test for phosphate Mixture + conc. HNO_3 , boil + excess of ammonium molybdate solution	—	—

C. Preparation of original solution (O.S) : Mixture is dissolved in 20 mL quantity of distilled water in a beaker, stir with glass rod to dissolve the mixture. Clear solution is obtained. Use this O.S. for further tests of acidic radicals.

D. Wet Tests for Anion (Acidic Radicals)

1. O.S. + AgNO_3	white ppt. insoluble in dil HNO_3	Halide may be (Cl^- , Br^- , I^-) present
If white ppt obtained in above test, perform following distinction test		
Distinction between Cl^- , Br^- and I^- : O.S. + dil. H_2SO_4 (till acidic) + chloroform + Cl_2 water (fresh) in excess, shake vigorously and observe the colour of chloroform layer carefully.	chloroform layer yellowish brown	Br^- present
2. O.S. + $\text{Ba}(\text{NO}_3)_2$ solution	white ppt insoluble in dil. HNO_3	SO_4^{2-} present
3. O.S. + dil. acetic acid + (Freshly prepared) FeSO_4 solution		
4. O.S. + diphenyl amine + conc. H_2SO_4		
5. O.S. + FeCl_3 solution		
6. O.S. + dil. acetic acid + CaCl_2		
7. O.S. + dil. H_2SO_4 + 2-3 drops of KMnO_4		

E. Confirmatory tests for Acidic Radicals

1. C.T. for first detected acidic radical

Test	Observation	Inference
1. 0.5 gm MnO_2 + conc. CH_3COOH and heat	Reddish brown gas turning starch paper yellowish brown starch iodine paper blue	Br^- confirmed
2. 0.5 g Cl_2 water + CHCl_3 shake well	chloroform layer acquired yellow colour	Br^- confirmed

2. C.T. for second detected acidic radical

Test	Observation	Inference
1. 0.5 g lead acetate solution	white ppt insoluble in dil. HNO_3 but soluble in hot ammonium acetate	SO_4^{2-} confirmed
2. 0.5 g BaCl_2	white ppt of BaSO_4	SO_4^{2-} confirmed

Result :

The given inorganic mixture no.2 contains following two anions (Acidic radicals)

- Br^- (Name of anion) (Bromide)
- SO_4^{2-} (Name of anion) (Sulphate)

Remark and sign of teacher: